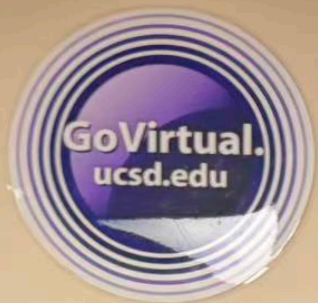
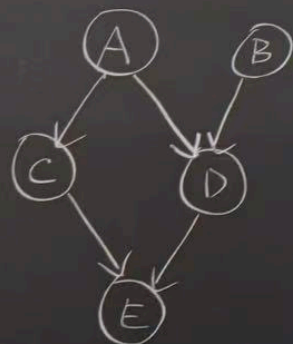


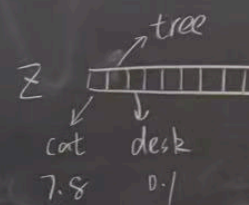
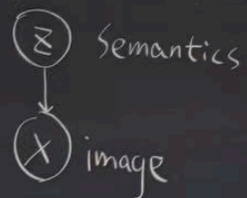


Bayesian network



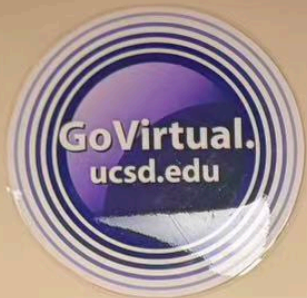
$$P(A, B, C, D, E) \\ = P(A) P(B) P(C|A) P(D|A, B) P(E|C, D)$$

Variational autoencoder (VAE)

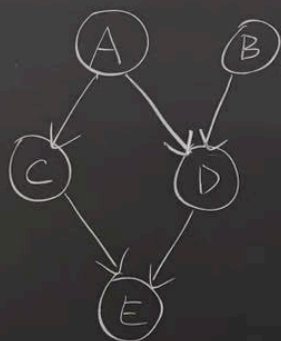


$$P(X|Z) \\ = N(\underline{x} | f(z), \text{diag}(\sigma^2))$$

$$p(x, z) = p(z) p(x|z)$$



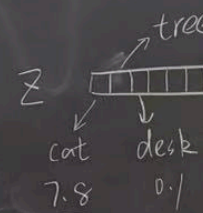
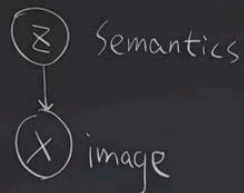
Bayesian network



$$P(A, B, C, D, E)$$

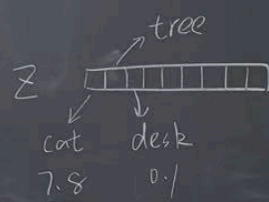
$$= P(A) P(B) P(C|A) P(D|A, B) P(E|C, D)$$

Variational autoencoder (VAE)

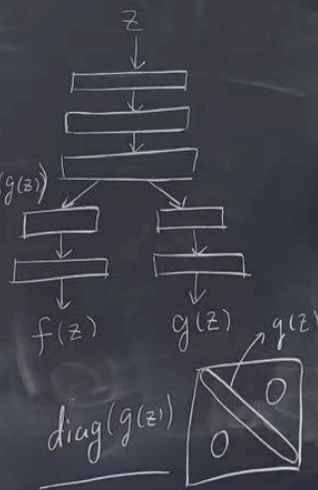


$$P(X|Z)$$
$$= N(\underline{x} | f(\underline{z}), \sigma)$$

$$p(x, z) = p(z) p(x|z)$$



$$z \sim N(\vec{0}, I)$$



B) $P(E|C, D)$

$$P(X|Z) = N(X | f(z), \text{diag}(g(z)))$$

$$p(x, z) = p(z) p(x|z)$$

① Model training (parameter learning)

② Generate data (sampling)

$$\hat{z} \sim N(\vec{0}, I)$$

$$\hat{x} \sim N(x | f(\hat{z}), \text{diag}(g(\hat{z})))$$

③ Data understanding (inference)

$$p(z|x) = \frac{p(z, x)}{p(x)}$$

$$p(x) = \int_z p(z, x) dz$$

NP-hard



minimize
Variational

arning)

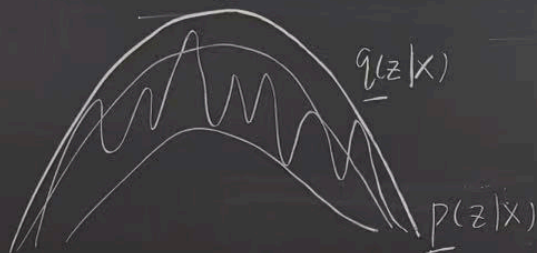
I)

$$f(\hat{z}), \text{diag}(q(\hat{z}))$$

erence)

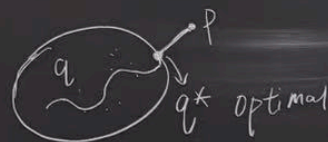
$$p(z, x) dz$$

NP-hard

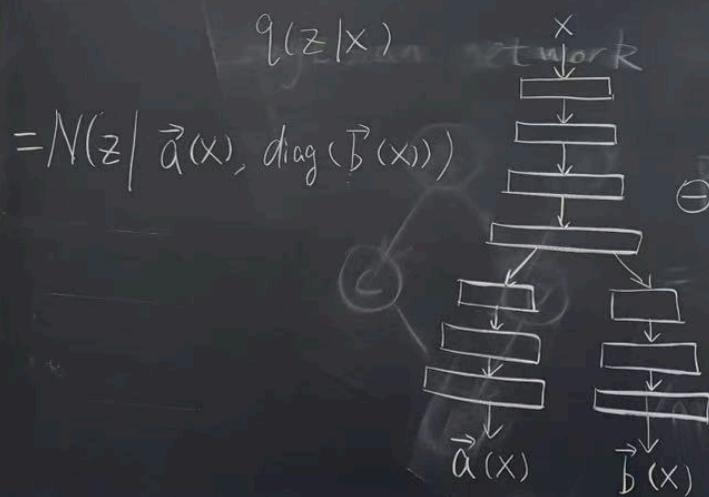


minimize the difference between $p(z|x)$ and $q(z|x)$.

Variational inference - optimization



er wen mu du
suo wen
qian yi yi
zhi jin
guo fang
jie bai
die dao
po lan
hou shang
tao pa xue liu



Kullback - Leibler Divergence
(KL)

$$\downarrow KL(q(z|x) || p(z|x)) \geq 0$$

$$= E_{q(z|x)} \left[\log \frac{q(z|x)}{p(z|x)} \right]$$

$$E_{q(z)} [f(z)]$$

ence

$$x) \geq 0$$

$$\min_{\Theta} KL(q_{\theta}(z|x) \parallel p(z|x)) = E_{q(z|x)} \left[\log \frac{q(z|x)}{p(z|x)} \right]$$

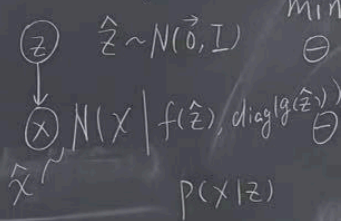


$$q_{\theta^*} \approx p(z|x)$$

diag(q_{θ^*})

Latent (parameter)

Generate image



$$\begin{aligned} &= E_{q(z|x)} [\log q(z|x) - \log p(z|x)] \\ &= E_{q(z|x)} [\log q(z|x)] - E_{q(z|x)} [\log p(z|x)] \end{aligned}$$

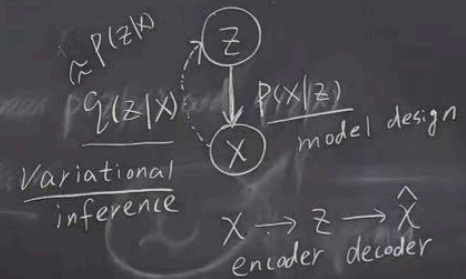
$$\begin{aligned} &= E_{q(z|x)} [\log p(z|x)] \\ &= E_{q(z|x)} \left[\log \frac{p(z|x)}{p(x)} \right] = E_{q(z|x)} [\log p(z|x)] - E_{q(z|x)} [\log p(x)] \end{aligned}$$

Generate image

$$\min_{\Theta} E_{q_{\Theta}(z|x)} [\log q_{\Theta}(z|x)] - E_{q_{\Theta}(z|x)} [\log p(z,x)] + \log p(x)$$

$\hat{z} \sim N(\vec{0}, I)$
 $\hat{x} \sim N(x | f(\hat{z}), \text{diag}(\sigma^2(\hat{z})))$
 $q_{\Theta^*}(z|x) \approx p(z|x)$
 $p(x|z)$
 $E_{q(z)} [f(z)]$
 $z \sim N(z | \mu, \sigma^2)$
 $f(z) = z^2 + 3z$

$\log p(z|x) - \log p(z)$
 $\log p(z|x) - E_{q(z|x)} [\log p(z|x)]$
 $\log \frac{p(z,x)}{p(x)} = E_{q(z|x)} [\log p(z,x)] - E_{q(z|x)} [\log p(x)]$



er wen mu du
 su wen
 gu yi yi
 zhi jin
 guo feng
 jie bai
 die dao
 po lan
 hou shang
 tai po xue lu